4-Person Poster Presentation Script

Project: A Defective Product Detection System Using Image Processing and Machine Learning

Target total time: approximately 6–7 minutes | **Target per person:** 90–105 seconds

How this script matches the assessment

Assessment requirement	How your presentation demonstrates it
Knowledge	Explains image processing, segmentation, feature extraction, ML classification, and evaluation.
Application	Connects the pipeline to manufacturing inspection and a possible conveyor-belt deployment.
Communication	Each speaker has a focused section, clear transition, and a concise technical explanation.
Project-based learning	Shows the complete Problem → Methodology → Implementation → Output/Results story.
Critical thinking	Discusses model selection, lighting, dataset quality, false positives/negatives, and limitations.
Collaboration	Four speakers cover connected stages and hand over logically to one another.
Industry relevance	Frames the system as an automated visual quality-control pipeline.

Timing rule: *Speak naturally, do not rush. The scripts are intentionally concise. If your actual speaking speed is slower, remove one short paragraph; if faster, pause briefly at the bold technical terms.*

Person 1: Problem, Motivation, Introduction & Objective

Approx. script length: *213 words* | ***Expected delivery:*** *about 90–105 seconds at a normal presentation pace*

Speaking script

Good morning everyone. Our project is **“A Defective Product Detection System Using Image Processing and Machine Learning.”** The main problem we are addressing is quality inspection in manufacturing. In a conventional setup, products are visually inspected by people. This can be time-consuming, repetitive, expensive at large scale, and affected by fatigue or human error. A small crack, scratch, missing part, or shape abnormality can sometimes be overlooked.

Our idea is to make this inspection more systematic by using a camera, image processing, and machine learning. The system takes a product image as input and analyses its visual characteristics to decide whether the product is defective or non-defective. The poster shows examples of four defect categories: **scratch, crack, missing part, and incorrect shape**, along with normal products.

The objective is therefore not simply to classify an image, but to build a practical inspection pipeline. We first improve and prepare the image, then isolate the relevant product region, extract useful visual information, detect abnormal characteristics, and finally classify the product. This makes the project relevant to real manufacturing because the same idea can support faster and more consistent quality checking.

With that problem and objective established, I will hand over to my teammate, who will explain how the captured image is processed step by step.

Where to point on the poster

Point to: Abstract → Introduction → Defect Detection Examples. Emphasize the practical manufacturing problem and the objective.

Transition

“That explains why the problem matters. Now we will show how the image is prepared before the system makes a decision.”

Person 2: Image Acquisition, Enhancement, Restoration, Preprocessing & Segmentation

Approx. script length: *206 words* | ***Expected delivery:*** *about 90–105 seconds at a normal presentation pace*

Speaking script

I will now explain the **image-processing workflow** shown in the centre of our poster. The first stage is **image acquisition**. A camera captures the product image, and these images form the dataset used for analysis. At this stage, consistent camera position and lighting are important because shadows, reflections, or poor focus can create patterns that may be mistaken for defects.

The second stage is **image enhancement**. We improve image quality by reducing noise and increasing useful contrast. The third stage is **image restoration**, which deals with distortions or degradation in the acquired image. The distinction is that enhancement improves useful visibility, while restoration attempts to recover information affected by degradation.

Next comes **preprocessing**. Images can be resized to a standard input size, normalized to a consistent numerical range, and filtered when required. Normalization, for example, can scale an 8-bit pixel value from 0–255 into a range such as 0–1.

The fifth stage is **image segmentation**. Here, we separate the product from its background and extract the region of interest. This is important because the model should learn from the product and its defects rather than irrelevant background information.

After preparing the image, we can move to extracting meaningful characteristics and making the final prediction.

Where to point on the poster

Point to workflow steps 1–5. Use the nut image to explain why background, lighting and preprocessing matter.

Transition

“Once the image is prepared and the region of interest is isolated, the next step is to extract information and classify the product.”

Person 3: Feature Extraction, Defect Detection, Classification & Proposed Model

Approx. script length: *211 words* | ***Expected delivery:*** *about 90–105 seconds at a normal presentation pace*

Speaking script

After preprocessing and segmentation, the next stage is to understand what information in the image can indicate a defect. Our poster groups these characteristics into **shape, colour, texture, and edge features**. Shape can include measurements such as area, perimeter, aspect ratio, or contour characteristics. Colour features can capture intensity or colour distributions. Texture features describe surface patterns, and edge features highlight sudden changes that may indicate boundaries, cracks, or other abnormalities.

These features support the **defect-detection** stage, where abnormal regions can be identified and compared with the expected characteristics of a normal product. The system then performs **classification**. The poster lists candidate machine-learning approaches including Decision Tree, Random Forest, SVM, and CNN or another deep-learning model. A traditional model generally uses extracted numerical features, while a CNN can learn image features automatically from training examples.

The important point is that the final classifier should be selected based on actual experimental results rather than simply assuming one algorithm is best. We would compare models using measures such as accuracy, precision, recall, F1-score, and also consider inference speed and computational requirements.

So, the proposed model follows a clear chain: **input image** → **preprocessing** → **feature extraction** → **defect detection** → **machine-learning classification** → **digital output.** The next speaker will explain that output, the applications, and the overall conclusion.

Where to point on the poster

Point to: Proposed Model → Feature Extraction → Defect Detection → Machine Learning Classification → Digital Output. Do not claim a final algorithm unless your team has actually implemented it.

Transition

“Now that we have seen how the prediction is made, the final speaker will explain the output, practical applications, limitations, and future scope.”

Person 4: Output, Applications, Conclusion & Future Scope

Approx. script length: *229 words* | ***Expected delivery:*** *about 90–105 seconds at a normal presentation pace*

Speaking script

I will conclude our presentation by explaining the output and practical value of the proposed system. The final decision shown on the poster is whether the product is **defective or non-defective**. The examples make the idea easy to interpret: a scratch, crack, missing part, or incorrect shape should lead to a defective decision, while a product matching the expected characteristics should be accepted.

In a real manufacturing environment, this pipeline could be connected to a conveyor-belt camera system. Each product could be photographed, processed, classified, and, if defective, flagged for human inspection or sent to an automated rejection mechanism. Possible applications include automotive components, metal parts, electronics, packaging, and other quality-control processes where visual defects matter.

There are also limitations. Performance depends on the quality and diversity of the dataset, camera conditions, lighting, product orientation, and the types of defects represented during training. A model can also produce false positives or false negatives, so evaluation on unseen test data is essential. For future work, we can add real-time inspection, more defect categories, defect localization, severity estimation, and edge deployment on industrial or embedded hardware.

To conclude, our project combines **image processing and machine learning** to make visual inspection faster, more consistent, and less dependent on repetitive manual checking. The key contribution is the complete problem-to-output workflow shown on our poster. Thank you. We are happy to take your questions.

Where to point on the poster

Point to: Output → Defect Detection Examples → Conclusion → bottom application/value strip. End confidently and invite questions.

Transition

“That concludes our presentation. Thank you. We are ready for your questions.”

Likely Questions After the Presentation

1. What is the main objective of the project?

Answer: To automate visual quality inspection by analysing product images and classifying products as defective or non-defective.

2. Why use image processing before machine learning?

Answer: It can reduce noise, improve contrast, standardize the input, and isolate the relevant product region. This can make the information given to a classifier more useful.

3. Why is segmentation important?

Answer: It separates the product or region of interest from the background, reducing irrelevant information and helping the system focus on the actual product.

4. What is the difference between enhancement and restoration?

Answer: Enhancement improves useful visibility or appearance, while restoration attempts to recover an image affected by a known or modeled degradation such as blur or distortion.

5. Why is lighting important?

Answer: Shadows and reflections can resemble defects. Controlled lighting or robust training helps the system distinguish real defects from illumination changes.

6. Why machine learning instead of fixed rules?

Answer: Fixed thresholds and rules can fail when defect appearance varies. Machine learning can learn more complex patterns from labelled examples.

7. Why might you choose a CNN?

Answer: CNNs are designed for image data and can learn spatial features such as edges, textures, and shapes directly from images.

8. Why might Random Forest or SVM be useful?

Answer: They can work well with structured features such as shape, colour, and texture, especially when the dataset or computational resources are limited.

9. How do you evaluate the model?

Answer: Use a held-out test set and metrics such as accuracy, precision, recall, F1-score, and a confusion matrix. In industrial inspection, recall for defective products can be especially important.

10. What is a false negative?

Answer: An actually defective product is predicted as non-defective. This can be serious because the defective product may pass quality control.

11. What is overfitting?

Answer: The model learns the training data too closely and performs poorly on unseen data. Data augmentation, regularization, dropout, early stopping, and proper train/validation/test separation can help.

12. What happens if a new defect type appears?

Answer: A supervised model may not reliably recognize a defect that was not represented in training. The dataset and defect classes would need to be expanded, or an anomaly-detection approach could be investigated.

13. Is this classification or detection?

Answer: Classification answers whether the product belongs to a class such as defective or non-defective. Detection/localization identifies where an abnormality is present. A complete system can combine both.

14. Can this be used in real time?

Answer: Potentially yes. Real-time performance depends on image resolution, preprocessing, model size, hardware, and the required production speed.

15. What is the biggest limitation?

Answer: The system is highly dependent on representative, high-quality data and consistent imaging conditions. It should be validated on unseen real-world samples before industrial deployment.

Team Final Checklist Before Presentation

■ Know the exact dataset source and total number of images.

■ Know the number of defective and non-defective samples.

■ Know the train/validation/test split, if used.

■ Know the exact final model you actually implemented. Do not claim that all listed algorithms were implemented unless they were.

■ Know the input image size and the main preprocessing steps.

■ Know which segmentation and feature-extraction techniques were actually used.

■ Know your measured accuracy, precision, recall and F1-score if experiments are complete.

■ Be able to explain one false positive and one false negative.

■ Be ready to explain why your chosen model is appropriate.

■ Be honest about limitations and future work.

■ All four members should know the complete pipeline, not only their own section.

■ Practice once with a timer. Aim for about 95 seconds per person, leaving a small buffer for handovers.

Golden one-sentence answer: “Our system captures a product image, improves and segments it, extracts or learns relevant visual features, detects abnormal characteristics, and uses a machine-learning/deep-learning classifier to decide whether the product is defective or non-defective.”